Escuela de
Ingeniería y Arquitectura
Universidad Zaragoza

MUSIC & AI

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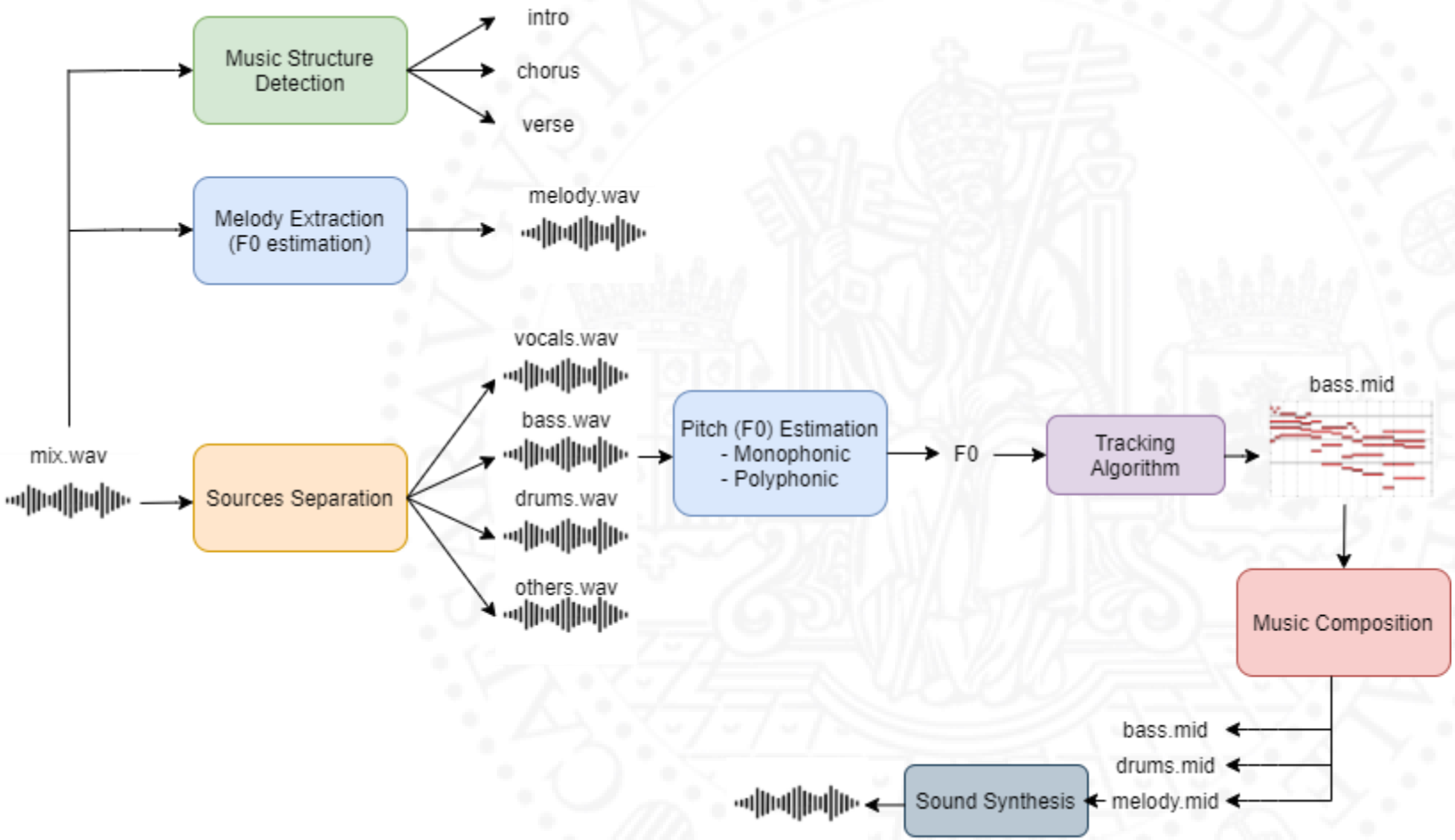
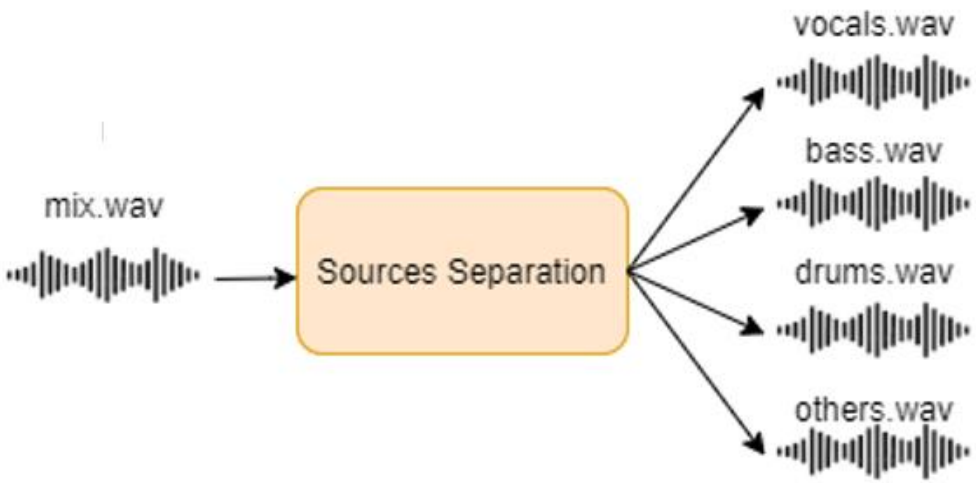
Introduction



Sources Separation

Separation of a set of source signals from a set of mixed signals, without the aid of information (or with very little information) about the source signals or the mixing process

- Inputs → Spectrograms
- Metrics → SDR, SIR, SAR

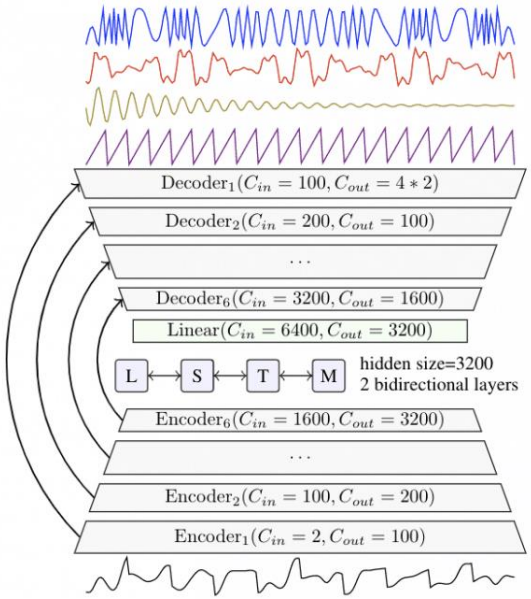


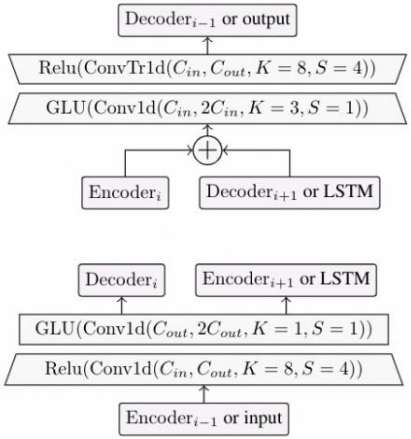
Sources Separation

1

"DEMUCS"
Défossez, A., Usunier, N., Bottou, L., & Bach, F. (2019). **Demucs: Deep Extractor for Music Sources with extra unlabeled data remixed.** arXiv preprint arXiv:1909.01174.

- **U-net**: Convolutional Autoencoder with skip connections
- **Train dataset**: MusDB18





Sources Separation

2

“SPLEETER”
Hennequin, R., Khlif, A., Voituret, F., & Moussallam, M. (2020). **Spleeter: a fast and efficient music source separation tool with pre-trained models.** Journal of Open Source Software, 5(50), 2154.

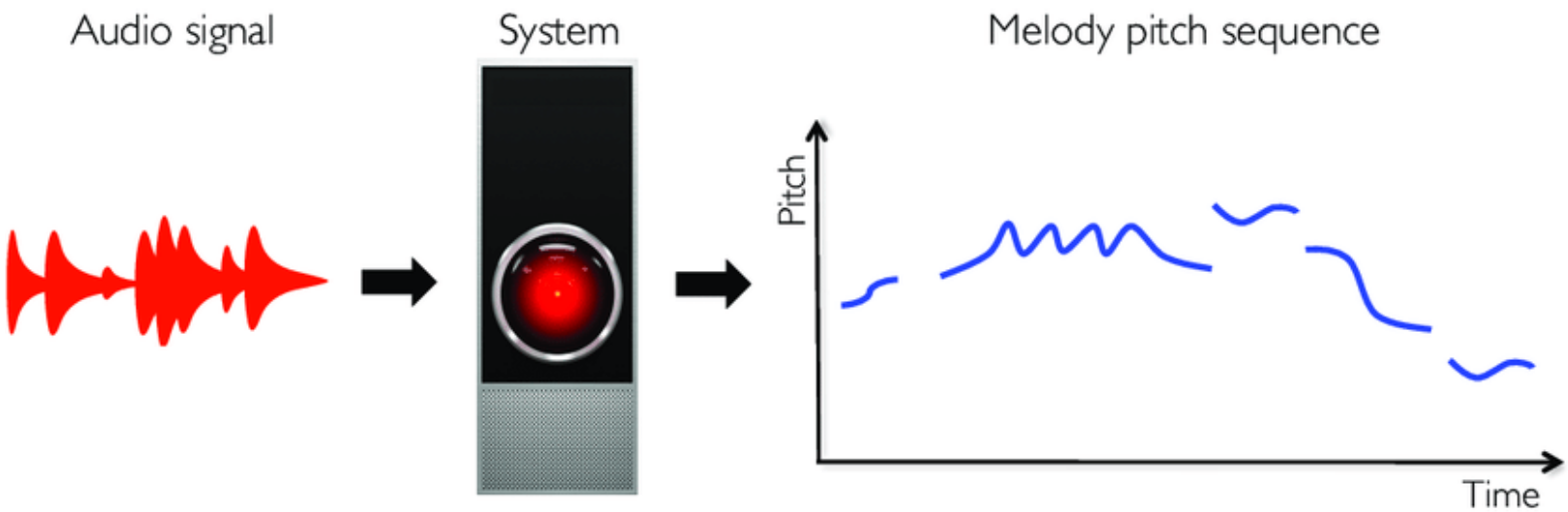
- **U-net**: Convolutional Autoencoder with skip connections
- **Train dataset**: private

  - Also separates piano from mix.wav

Pitch Estimation: Melody Extraction

Estimate the **fundamental frequency** corresponding to the pitch of the **predominant melodic** line of a piece of **polyphonic** music.

- Inputs → Constant Q-Transform (CQT)
- Metrics → Voicing Recall Rate, Voicing False Alarm Rate, Raw Pitch Accuracy, Raw Chroma Accuracy, Overall Accuracy



Justin Salamon web

Pitch Estimation: Monophonic (F0)

- Inputs → Constant Q-Transform (CQT)
- Metrics → Raw pitch accuracy (RPA) and Raw chroma accuracy (RCA)

1

"SWIPE"
Camacho, A. and Harris, J. G. **A sawtooth waveform inspired pitch estimator for speech and music.** The Journal of the Acoustical Society of America, 124(3):1638-1652, 2008.

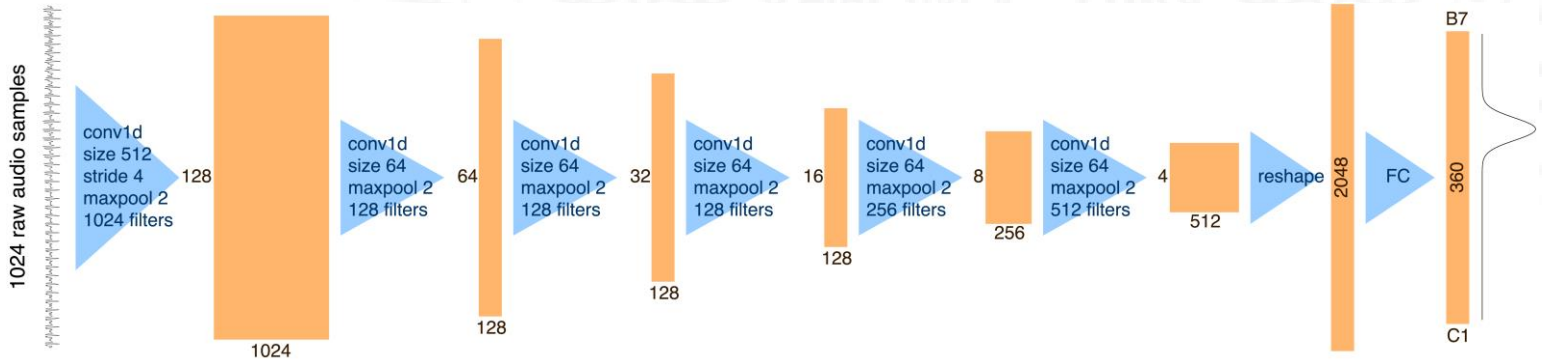
No Deep Learning

Pitch Estimation: Monophonic (F0)

2

"CREPE"
Kim, J. W., Salamon, J., Li, P., & Bello, J. P. (2018, April). **CREPE: A convolutional representation for pitch estimation.** In 2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 161-165). IEEE.

- **CNN** with 487k params (*tiny*) and 22.2M (*full*)
- **Training Datasets:** MedleyDB, Bach10, RCW-Synth and NSynth

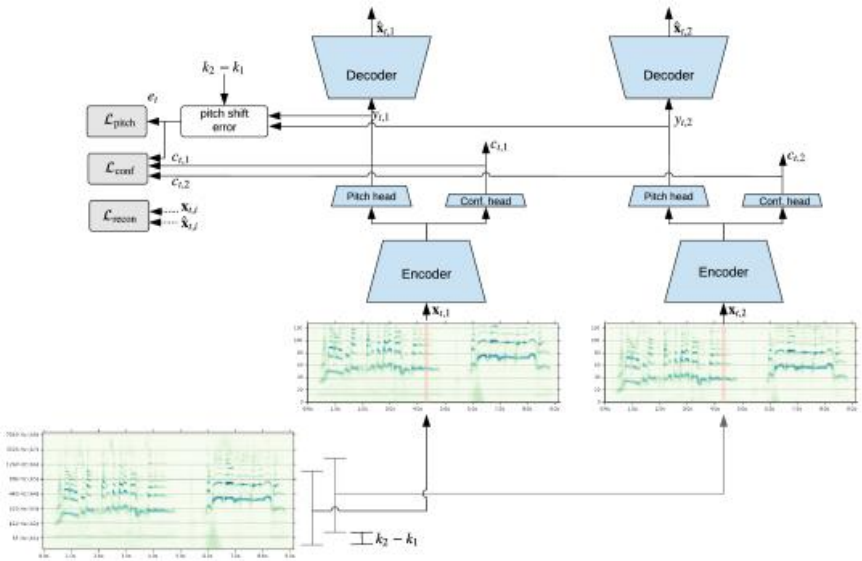


Pitch Estimation: Monophonic (F0)

3

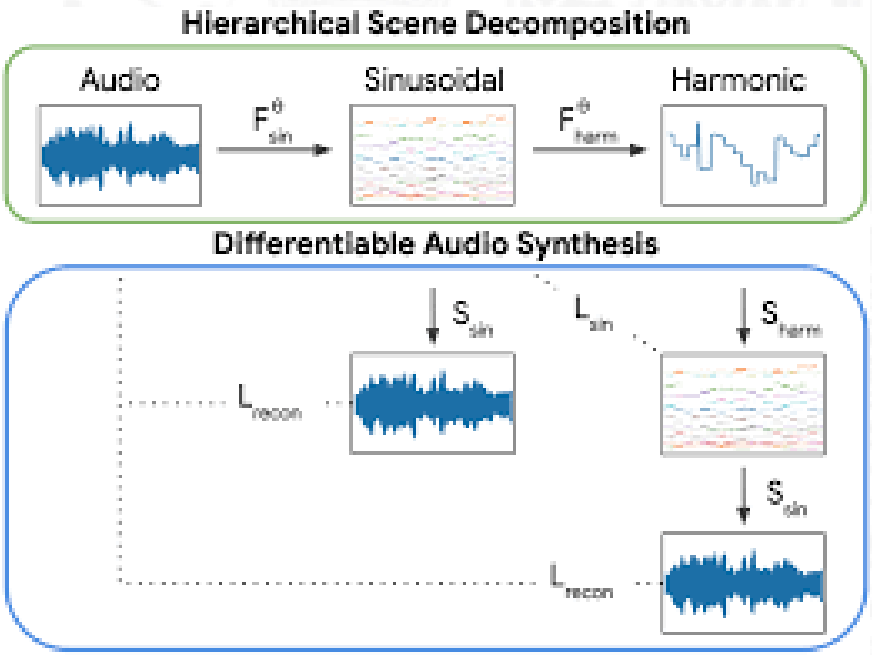
"SPICE"
Gfeller, B., Frank, C., Roblek, D., Sharifi, M., Tagliasacchi, M., & Velimirović, M. (2020). **SPICE: Self-supervised Pitch Estimation.** IEEE/ACM Transactions on Audio, Speech, and Language Processing, 28, 1118-1128.

- Inputs: CQTs
- Self-Supervised **Autoencoder** with 2.38M params.
- **Training Datasets**: MIR-1k and "singing voices"
- **Test Datasets**: MedleyDB and MIR-1k



Pitch Estimation: Monophonic (F0)

4 Engel, J., Swavely, R., Hantrakul, L. H., Roberts, A., & Hawthorne, C. (2020). Self-supervised Pitch Detection by Inverse Audio Synthesis.

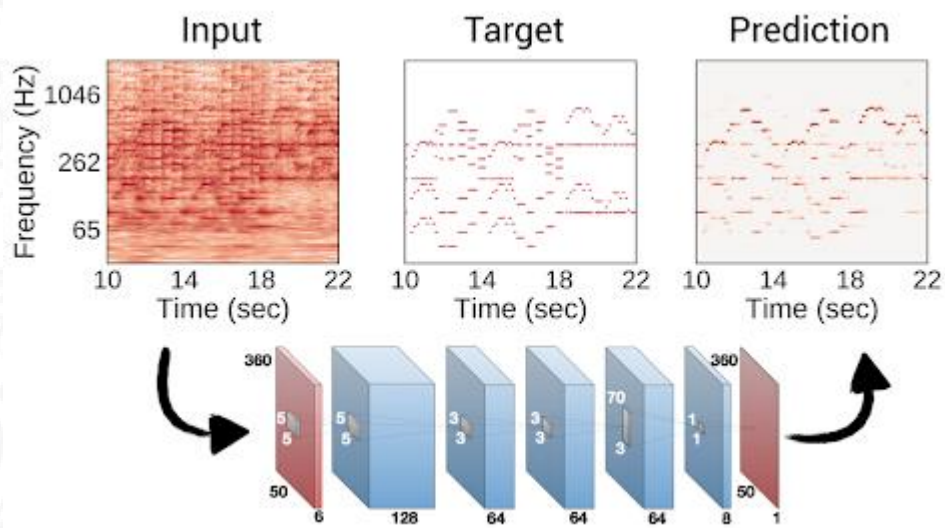


Pitch Estimation: Polyphonic (multi-F0)

1

"DEEP SALIENCE"
Bittner, R. M., McFee, B., Salamon, J., Li, P., & Bello, J. P. (2017, October). **Deep Salience Representations for F0 Estimation in Polyphonic Music.** In ISMIR (pp. 63-70).

- **Inputs**: HCQTs
- CNN
- **Training Datasets:** MedleyDB + "pop songs"
- **Test Datasets:** Bach10, Su datasets, MedleyDB



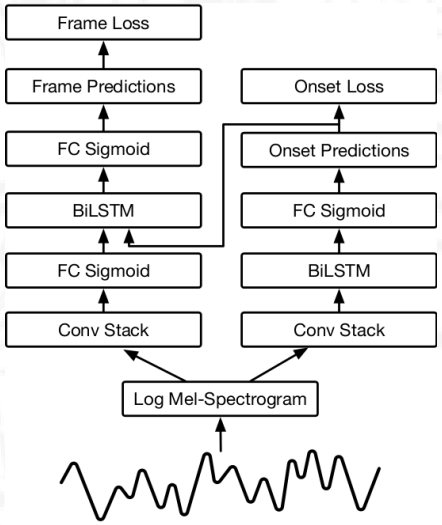
Music Transcription

- Metrics → Precision, Recall and F-measure for Onsets, Offsets and Overall

1

"ONSETS AND FRAMES"
Hawthorne, C., Elsen, E., Song, J., Roberts, A., Simon, I., Raffel, C., ... & Eck, D. (2017). **Onsets and frames: Dual-objective piano transcription.** arXiv preprint arXiv:1710.11153.

- **Inputs**: Log-Mel Spectrogram
- CNNs and Bi-LSTMs
- **Training Datasets**: Maestro (piano), E-GMD (drums)
- **Test Dataset**: MAPS dataset

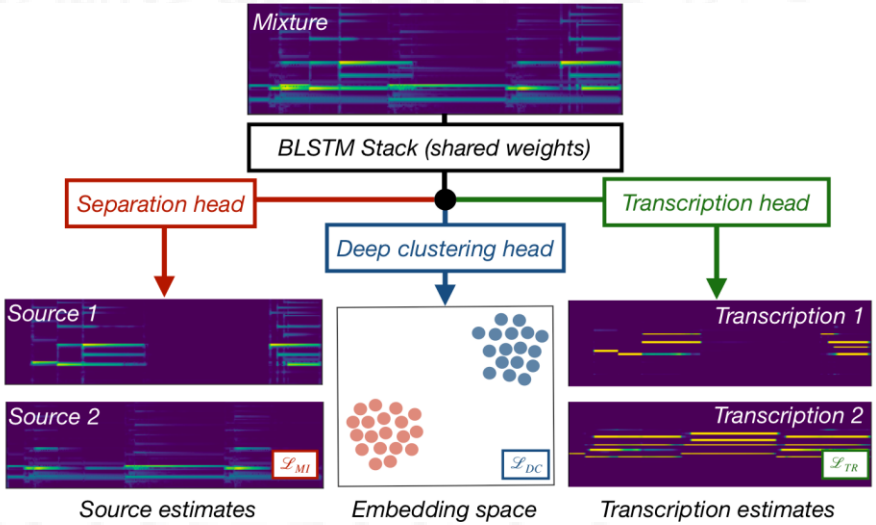


Sources Separation & Music Transcription

1

"Cerberus"
Manilow, E., Seetharaman, P., & Pardo, B. (2020, May). Simultaneous Separation and Transcription of Mixtures with Multiple Polyphonic and Percussive Instruments. In *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 771-775). IEEE.

- **Inputs**: Spectrogram
- Bi-LSTMs
- **Training Datasets**: Slakh2100
- **Test Dataset**: MAPS dataset

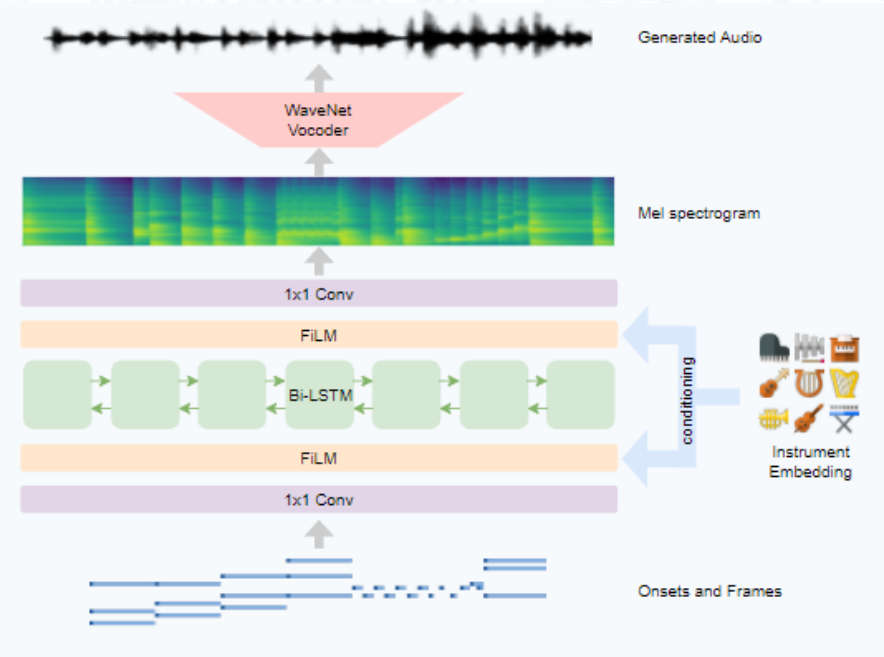


Sound Synthesis

1

Kim, J. W., Bittner, R., Kumar, A., & Bello, J. P. (2019, May). **Neural music synthesis for flexible timbre control**. In *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 176-180). IEEE.

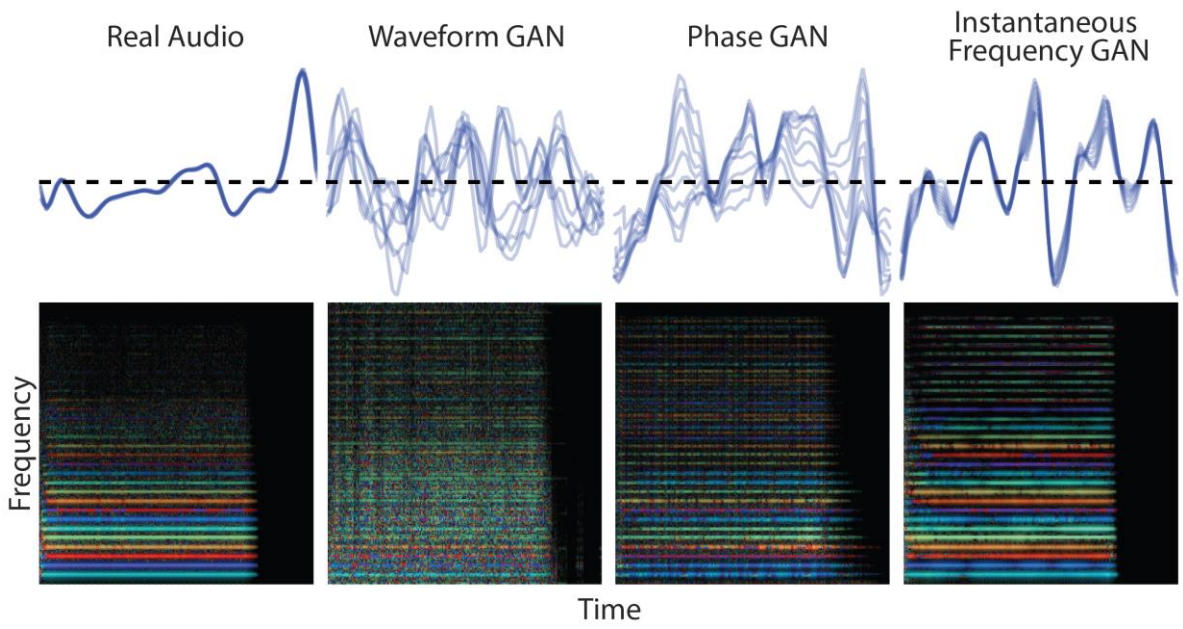
- **Inputs**: Note sequence
- Bi-LSTMs and WaveNet
- **Training Datasets**: created with FluidSynth and Soundfonts from "Classical Piano MIDI page"



Sound Synthesis

2

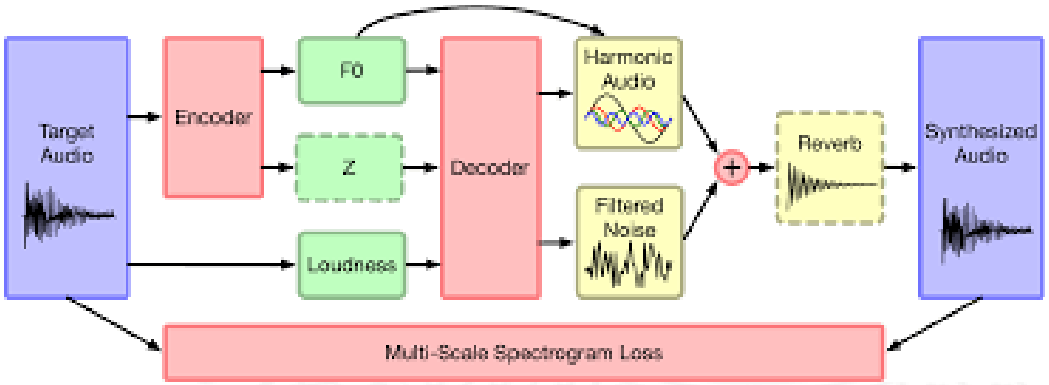
"GANSynth"
Engel, J., Agrawal, K. K., Chen, S., Gulrajani, I., Donahue, C., & Roberts, A. (2019). Gansynth: Adversarial neural audio synthesis. *arXiv preprint arXiv:1902.08710*.



Sound Synthesis

3

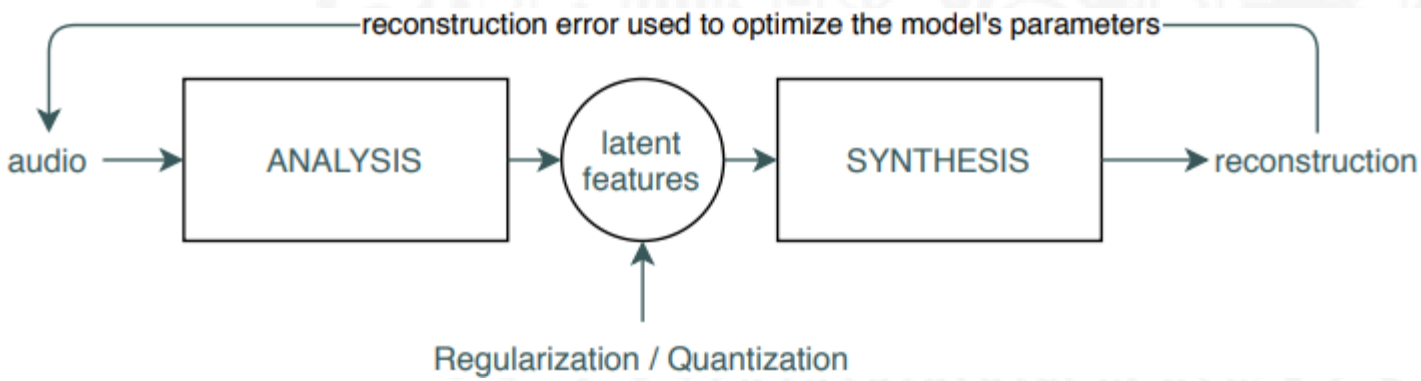
“DDSP”
Engel, J., Hantrakul, L., Gu, C., & Roberts, A. (2020). **Ddsp: Differentiable digital signal processing.** arXiv preprint arXiv:2001.04643.



Sound Synthesis

4

"Timbre Latent Space"
Caillon, A., Bitton, A., Gatinet, B., & Esling, P. (2020). Timbre latent space: exploration and creative aspects. *arXiv preprint arXiv:2008.01370*.



Music Structure Detection

Identify and extract the components of the music structure such as the intro, chorus, verse…from the signal.

- Inputs → Self-Similarity Matrix or Self-Similarity Lag Matrix
- Metrics → Precision, Recall, F-measure

1 Serra, J., Müller, M., Grosche, P., & Arcos, J. L. (2014). **Unsupervised music structure annotation by time series structure features and segment similarity.** IEEE Transactions on Multimedia, 16(5), 1229-1240.

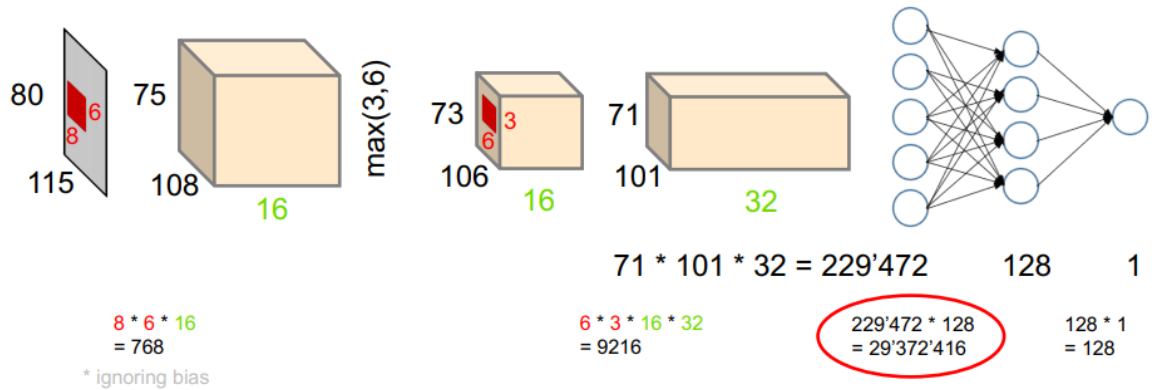
No Deep Learning

Music Structure Detection

2

Grill, T., & Schluter, J. (2015, August). **Music boundary detection using neural networks on spectrograms and self-similarity lag matrices.** In 2015 23rd European Signal Processing Conference (EUSIPCO) (pp. 1296-1300). IEEE.

- **Inputs**: SSLM
- CNN
- **Training Datasets**: private
- **Test Datasets**: SALAMI 2.0



Music Generation

Generate new music

The input of these models is normally a note sequence.

1

“WaveNet”
Oord, A. V. D., Dieleman, S., Zen, H., Simonyan, K., Vinyals, O., Graves, A., ... & Kavukcuoglu, K. (2016). **Wavenet: A generative model for raw audio**. *arXiv preprint arXiv:1609.03499*.

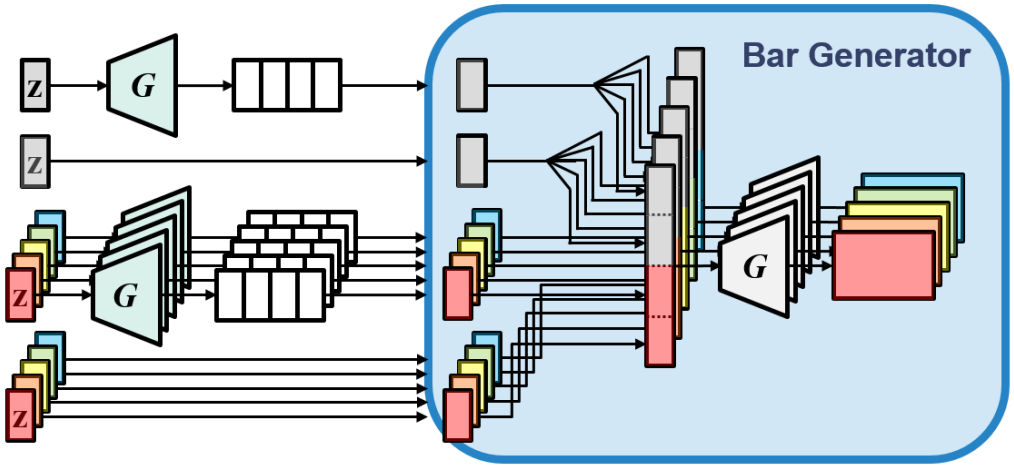
Not designed specifically for music

Music Generation

2

"MuseGAN"
Dong, H. W., Hsiao, W. Y., Yang, L. C., & Yang, Y. H. (2017). **Musegan: Multi-track sequential generative adversarial networks for symbolic music generation and accompaniment**. *arXiv preprint arXiv:1709.06298*.

- GAN
- **Training Datasets**:
  Lakh Pianoroll Dataset

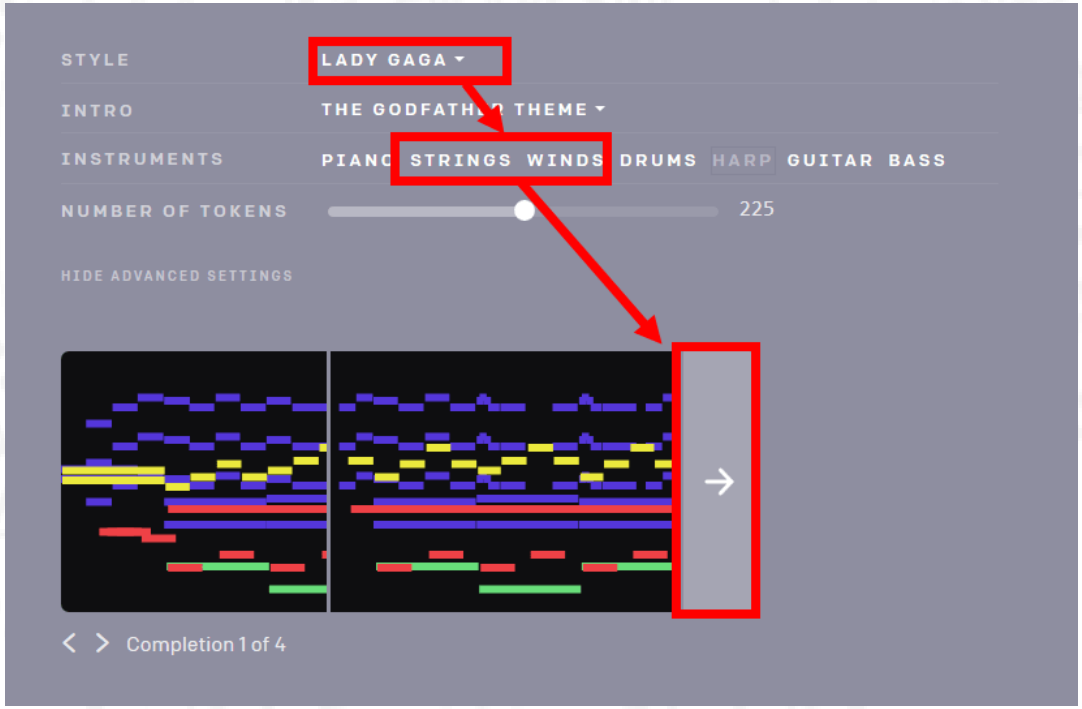


Music Generation

2

“MuseNet”
(2019, OpenAI)

- GAN

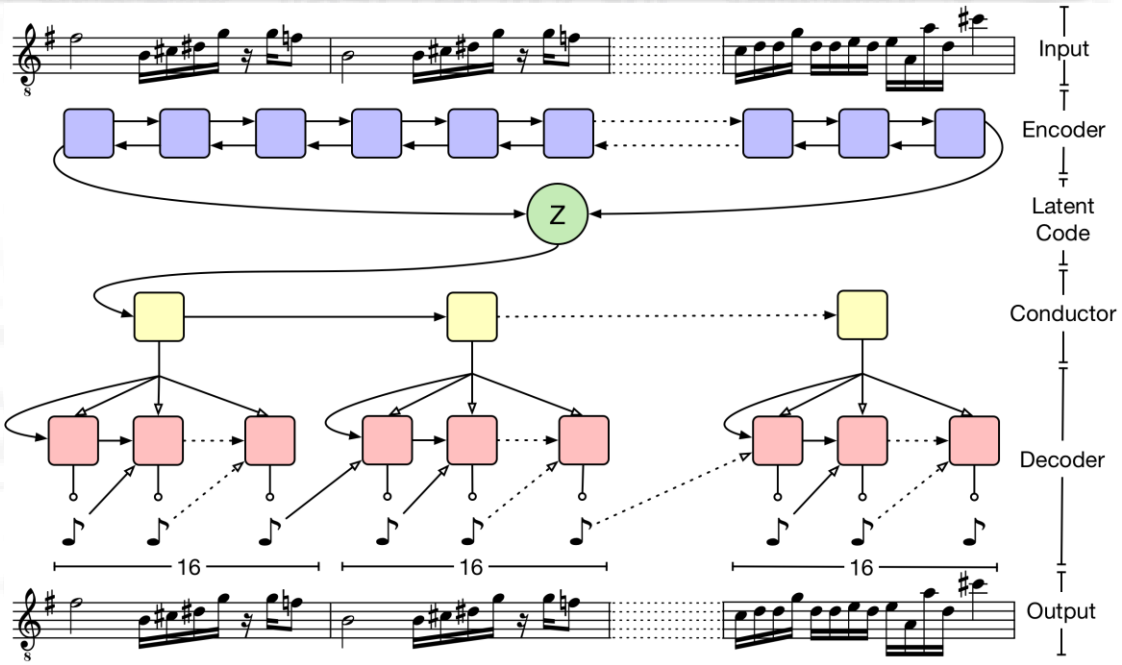


Music Composition

3

"MusicVAE"
Roberts, A., Engel, J., Raffel, C., Hawthorne, C., & Eck, D. (2018). **A hierarchical latent vector model for learning long-term structure in music.** arXiv preprint arXiv:1803.05428.

- Variational Autoencoder
- **Training Datasets**:

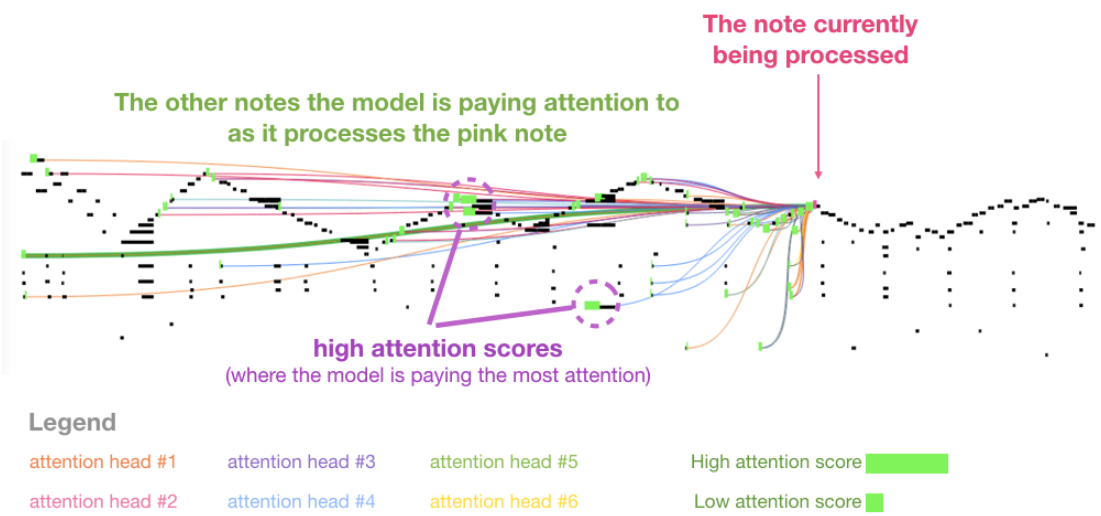


Music Composition

4

“Music Transformer”
Huang, C. Z. A., Vaswani, A., Uszkoreit, J., Simon, I., Hawthorne, C., Shazeer, N., ... & Eck, D. (2018, September). Music transformer: Generating music with long-term structure. In *International Conference on Learning Representations*.

- Transformer (attention mechanism)
- **Training Datasets**:

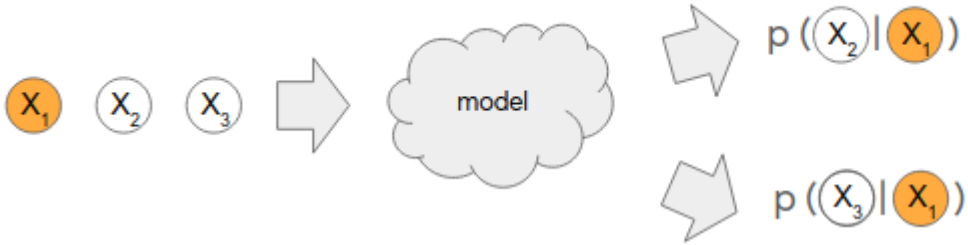


Music Generation: Accompaniment

1

"CocoNet"
Huang, C. Z. A., Hawthorne, C., Roberts, A., Dinculescu, M., Wexler, J., Hong, L., & Howcroft, J. (2019). **The bach doodle: Approachable music composition with machine learning at scale**. *arXiv preprint arXiv:1907.06637*.

- Gibbs sampling



$$p_{1,2,3}(\mathbf{x}) = p(x_1)\, p(x_2 | x_1)\, p(x_3 | x_1, x_2)$$

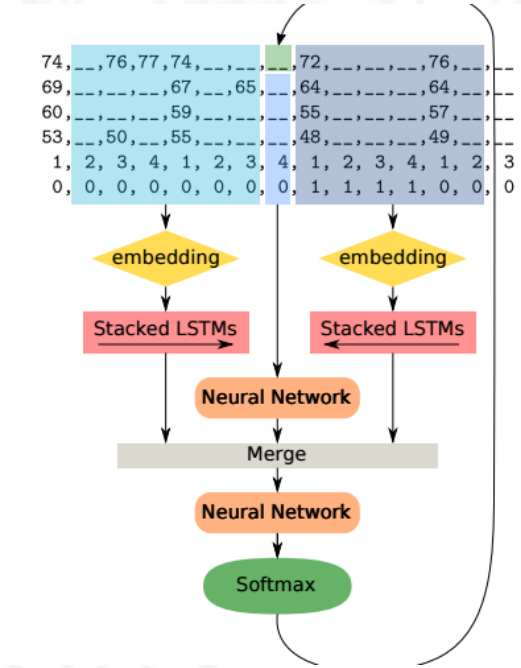
$$p_{1,3,2}(\mathbf{x}) = p(x_1)\, p(x_3 | x_1)\, p(x_2 | x_1, x_3)$$

Music Generation: Accompaniment

2

“DeepBach”
Hadjeres, G., Pachet, F., & Nielsen, F. (2017, July). **Deepbach: a steerable model for bach chorales generation**. In *International Conference on Machine Learning* (pp. 1362-1371). PMLR.

- LSTMs, Gibbs sampling
- **Training Datasets**: Bach Chorale Harmonizations from Music21



Datasets

	Dataset	Files	Possible tasks
1	Slakh2100	mix.flac mix.mid separated_stems.flac separated_stems.mid	Sources Separation Pitch Estimation
2	MedleyDB	mix.wav stems.wav F0_annotations	Sources Separation Melody Extraction Pitch Estimation
3	Salami 2.0	structure_annorations.txt songs.mp3	Music Structure Detection

Datasets

	Dataset	Files	Possible tasks
4	Lakh Midi Dataset	songs.mid	Music Generation Sound Synthesis Pitch Estimation (after synthesize)
5	RCW Music Databse		Music Structure Detection
6	E-GMD (Expanded Groove MIDI)	drums.mid drums.wav	Drums Transcription Drums Generation
7	Maestro	piano.mid piano.wav	Piano Transcription Piano Generation

Datasets

	Dataset	Files	Possible tasks
8	NSynth		Sound Synthesis Pitch Estimation
9	MIR-1k		Pitch Estimation
10	Bach10		Pitch Estimation
11	MAPS		Music Transcription